WORKED SOLUTION

STEP II 2013, Question 1

Pure • Chords of the Logarithm

STEP II 2013, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q1

This question has a full guided treatment in *Beyond STEP*, Section 3.5, π^e *or* e^π*?*: https://beyondstep.keithmcnulty.org/ch03/pi-e-or-e-pi.html

The key idea

Every inequality here is a comparison of p^q with q^p. Taking logarithms and dividing by pq makes it a comparison of $\frac{\ln p}{p}$ with $\frac{\ln q}{q}$, and $\frac{\ln x}{x}$ is the gradient of the line from the origin to the point $(x, \ln x)$ on the curve. So everything is about lines and chords meeting $y = \ln x$. The facts we need are that the curve is concave (it bends downwards) and that the tangent from the origin touches it at $x = \mathrm{e}$.

Solution

Part (i)

Suppose $y = mx$ touches $y = \ln x$ at $x = x_0$. The gradients agree and the curves meet, so

$$m = \frac{1}{x_0}, \qquad mx_0 = \ln x_0.$$

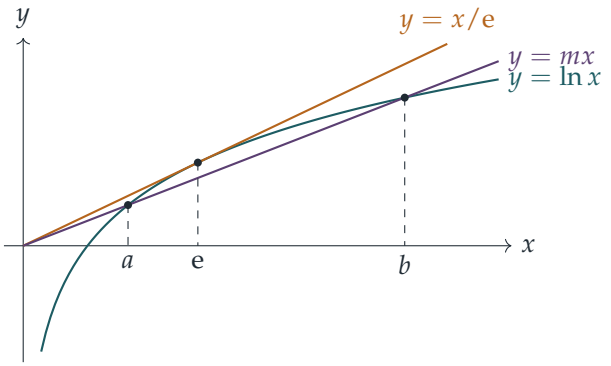
The first equation gives $mx_0 = 1$, so $\ln x_0 = 1$, $x_0 = \mathrm{e}$ and

$$m = \frac{1}{\mathrm{e}}.$$

Now suppose $y = mx$ meets the curve at $x = a$ and $x = b$. Then $\ln a = ma$ and $\ln b = mb$, so (as $a, b > 0$)

$$\frac{\ln a}{a} = m = \frac{\ln b}{b} \implies b \ln a = a \ln b \implies \ln\left(a^b\right) = \ln(b^a),$$

and since ln is one-to-one, $a^b = b^a$, as required.



For the sketch: if $m \leq 0$ the line meets the curve at most once, since $\ln x - mx$ is then strictly increasing. If $m \geq \frac{1}{\mathrm{e}}$ the line is the tangent or lies above it for $x > 0$, and the concave curve lies below its tangent $y = x/\mathrm{e}$, touching only at e, so again there are not two meeting points. So $0 < m < \frac{1}{\mathrm{e}}$, and the line $y = mx$ lies below the tangent, as in the figure. At $x = \mathrm{e}$ the curve is at height 1 and the line at height $m\mathrm{e} < 1$, so the curve is above the line there. Because the curve is concave and the line is straight, $\ln x - mx$ is concave. The set where it is positive is therefore a single interval, and its ends are the two meeting points a and b. That interval contains e, so $a < \mathrm{e} < b$.

Part (ii)

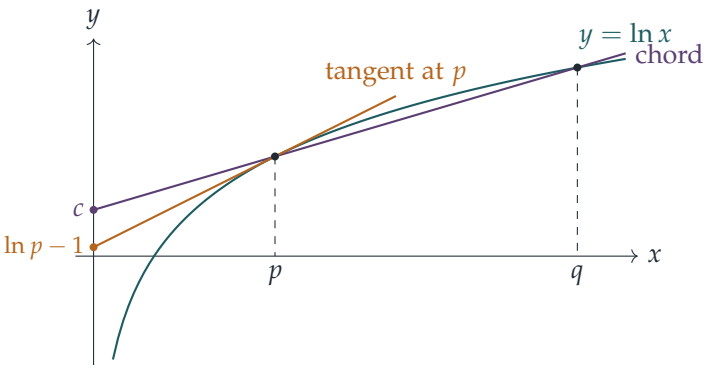
Here $\ln p = mp + c$ and $\ln q = mq + c$, with $0 < p < q$. Dividing by p and by q,

$$\frac{\ln p}{p} = m + \frac{c}{p}, \qquad \frac{\ln q}{q} = m + \frac{c}{q}.$$

Since $c > 0$ and $\frac{1}{p} > \frac{1}{q}$, we have $\frac{c}{p} > \frac{c}{q}$, so $\frac{\ln p}{p} > \frac{\ln q}{q}$. Multiplying by $pq > 0$ gives $q \ln p > p \ln q$, that is $\ln(p^q) > \ln(q^p)$, and so $p^q > q^p$.

In the picture, the lines from the origin to $(p, \ln p)$ and to $(q, \ln q)$ have gradients $\frac{\ln p}{p}$ and $\frac{\ln q}{q}$. If the chord crosses the y-axis above the origin, the nearer point subtends the steeper line from O. (The sign of c decides which way round the inequality goes. We use this again in part (iv).)

Part (iii)



Let the chord through $P = (p, \ln p)$ and $Q = (q, \ln q)$ have gradient s and y-intercept c, so $c = \ln p - sp$. Because the curve is concave, the chord from P to a point further right is less steep than the tangent at P, so

$$s < \frac{1}{p}.$$

(Or, by the mean value theorem, $s = \frac{1}{\xi}$ for some $\xi \in (p, q)$, and $\frac{1}{\xi} < \frac{1}{p}$.) The tangent at P is $y = \ln p + \frac{1}{p}(x - p)$, which meets the y-axis at $\ln p - 1$. The chord and the tangent both pass through P, and the chord is less steep. So moving left from P to $x = 0$, the chord stays above the tangent:

$$c = \ln p - sp > \ln p - 1 \geq 0,$$

using $p \geq \mathrm{e}$ at the last step. So the chord meets the y-axis at a positive value of y.

? **Ask yourself** How does this settle π^{e} against e^{π}?

Take $p = \mathrm{e}$ and $q = \pi$, which satisfy $\mathrm{e} \leq p < q$. The chord has $c > 0$, so part (ii) applies and gives $p^q > q^p$:

$$\mathrm{e}^{\pi} > \pi^{\mathrm{e}}.$$

Part (iv)

Now the chord through $(p, \ln p)$ and $(q, \ln q)$ has gradient $\frac{1}{\mathrm{e}}$, so it is $y = \frac{x}{\mathrm{e}} + c$ with

$$c = \ln p - \frac{p}{\mathrm{e}}.$$

The line $y = \frac{x}{\mathrm{e}}$ is the tangent at $x = \mathrm{e}$ from part (i), and the concave curve lies on or below it: $\ln x \leq \frac{x}{\mathrm{e}}$, with equality only at $x = \mathrm{e}$. So $c \leq 0$, with $c = 0$ only if $p = \mathrm{e}$. But $c = 0$ would make the chord the tangent itself, and the tangent meets the curve only at e, not at two points $p < q$. Hence

$$c < 0.$$

The chord is parallel to the tangent and lies below it. Now repeat the argument of part (ii) with the sign reversed:

$$\frac{\ln p}{p} = \frac{1}{\mathrm{e}} + \frac{c}{p}, \qquad \frac{\ln q}{q} = \frac{1}{\mathrm{e}} + \frac{c}{q}.$$

Since $c < 0$ and $\frac{1}{p} > \frac{1}{q}$, we get $\frac{c}{p} < \frac{c}{q}$, so $\frac{\ln p}{p} < \frac{\ln q}{q}$. Multiplying by $pq > 0$ gives $q \ln p < p \ln q$, and therefore

$$q^p > p^q.$$

(As in part (i), concavity also forces $p < \mathrm{e} < q$ here, but the argument does not need it.)

Checks

- $\mathrm{e}^\pi \approx 23.14$ and $\pi^\mathrm{e} \approx 22.46$, consistent with part (iii).
- Part (i): the pair $a = 2$, $b = 4$ has $2^4 = 4^2$, and indeed $\frac{\ln 2}{2} = \frac{\ln 4}{4}$ and $2 < \mathrm{e} < 4$.
- Part (iv): with $p = 1$, the chord gradient $\frac{\ln q}{q-1} = \frac{1}{\mathrm{e}}$ gives $q \approx 5.76$. Then $q^p \approx 5.76 > 1 = p^q$.
- Parts (ii) and (iii) were also confirmed by a computer check on many random chords.

Common slips

- Getting $q \ln p > p \ln q$ the wrong way round. Write out $\frac{\ln p}{p} > \frac{\ln q}{q}$ explicitly before multiplying by pq.
- In part (iii), stopping at "the curve is concave" without explaining why the intercept is positive. The tangent at p meets the axis at $\ln p - 1 \geq 0$, and the chord crosses above it.
- In part (iv), forgetting to rule out $c = 0$, which is needed for the strict inequality.

Beyond STEP

The function $\frac{\ln x}{x}$ has its maximum at e, and this explains the famous pair $2^4 = 4^2$. The positive solutions of $x^y = y^x$ with $x \neq y$ form a curve, which can be parametrised as $x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. More generally, "a chord of a concave function lies below the curve and above the tangents" is the basic inequality of convexity. It underlies Jensen's inequality and the AM–GM inequality. Bartle and Sherbert, *Introduction to Real Analysis*, covers the mean value theorem and convex functions carefully.

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